Sikkim E-Waste Signature Far Exceeds State & Global Background Levels

Comparative analysis of key e-waste markers in Sikkim soil/sediment samples, FY 2026

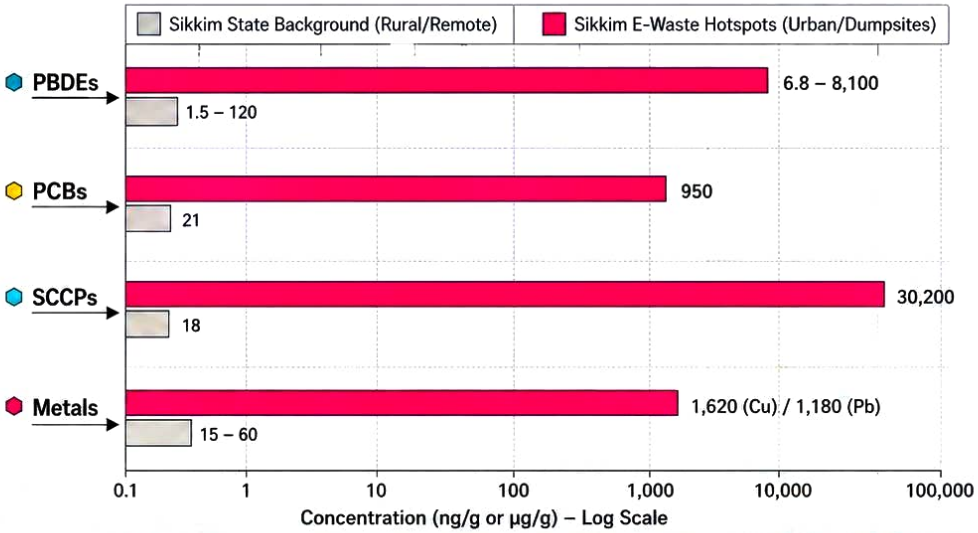
CLASSIFICATION: DIAGNOSTIC REPORT

PBDEs (Polybrominated Diphenyl Ethers)
Sikkim hotspots record 6.8 to 8,100 ng/g, vastly higher than state background (1.5 to 120 ng/g).

PCBs (Polychlorinated Biphenyls)
E-waste site soils reach up to 950 ng/g, far outpacing state background levels of 21 ng/g.

SCCPs (Short Chain Chlorinated Paraffins)
Extreme localized spikes hit 30,200 ng/g, compared to state background of 18 ng/g.

Metals (Heavy Metals)
Copper (1,620 µg/g) and Lead (1,180 µg/g) significantly exceed natural background levels in Sikkim soils.



Source: Sikkim State Pollution Control Board, E-Waste Monitoring Program, FY 2026
Baseline: Remote/Rural Soil & Sediment Averages (n=120)
Method: SOP-Based Sampling & Analysis (US EPA 3540C / 8270E / 6010D)
Note: Values indicate ranges/averages from sampled e-waste hotspots across Sikkim.

Key Takeaway:
E-waste hotspots in Sikkim show contamination levels several orders of magnitude higher than natural state background and global averages—urgent remediation & source control needed.

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